Unit ID: 679a14d7aaf593b2823f24b4

Unit 3 - Designing Solutions

PDF Documents

PDF Document 1: 17381511125495-Chapter 9 - Unit 3 - Designing Solutions.pdf

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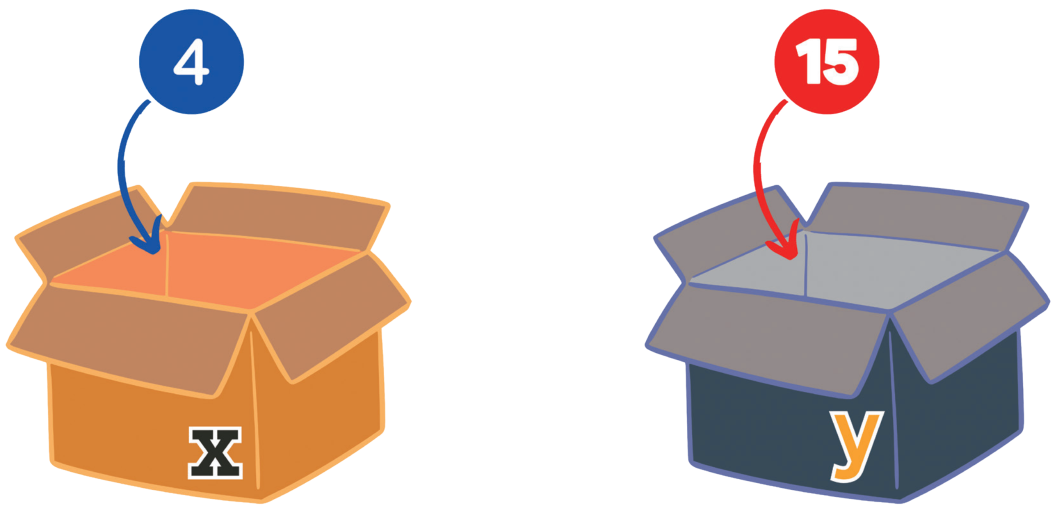
Update #1 - 2025-05-16 15:21:09

UNIT 3 DESIGNING SOLUTIONS

Programming Concepts

In mathematics, variables such as x and y are used to represent unknown values that can change. For instance, in the equation $x + 2 = 5$, x is a placeholder for a value we need to discover. Once we solve the equation, we find that x equals 3.

✓ VARIABLES

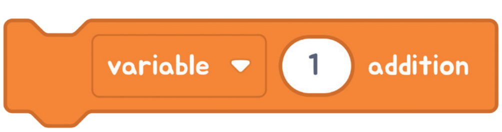


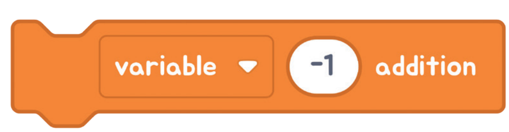
Similarly, in programming, a variable is a name assigned to a storage area in the computer's memory where data can be stored, modified, and retrieved later. Just like in math, the value of a programming variable can change as the program runs. For example, if we set a variable x in a program with $x = 5$, we can later change it to $x = 10$ based on some conditions or calculations performed during the execution of the program.

EXPLANATION WITH CODE EXAMPLE

INCREMENT AND DECREMENT OPERATORS

Increment and decrement operators are used to adjust a variable's value by a fixed amount, typically 1, which simplifies coding in repetitive tasks. These operators are especially useful in loops for iterating over sequences.



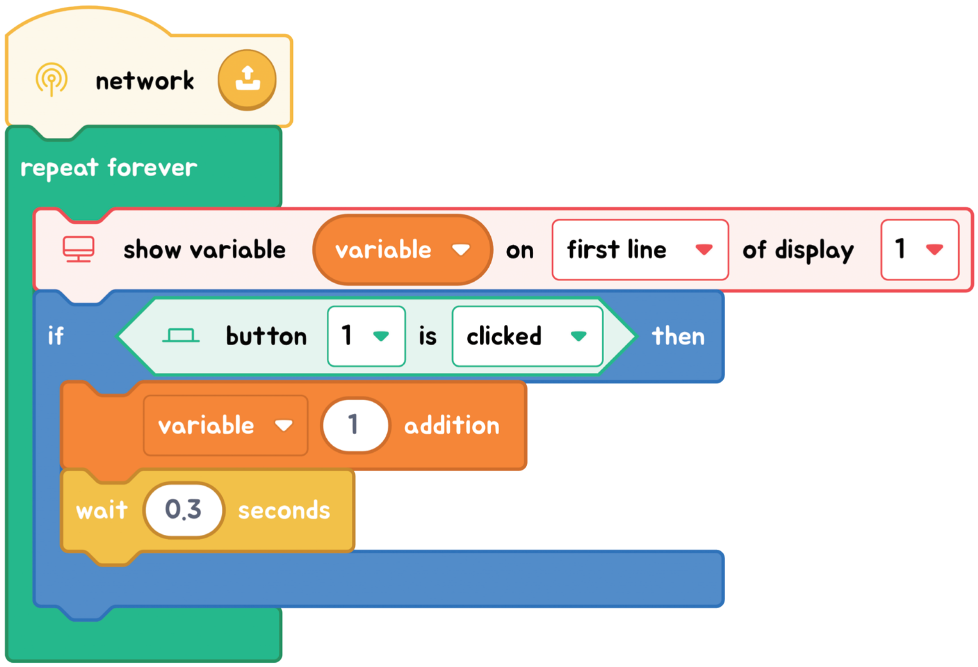


- The block on the left, known as the increment operator, increases the variable's value by 1.
- The block on the right, known as the decrement operator, decreases the variable's value by 1.

HOW ARE VARIABLES USED?

Variables make it easier to update and manage your code. Instead of repeatedly changing the same piece of information everywhere it appears in your program, you can use a variable. This means you only need to update the value in one place, and every part of your program that uses this variable will automatically use the new value.

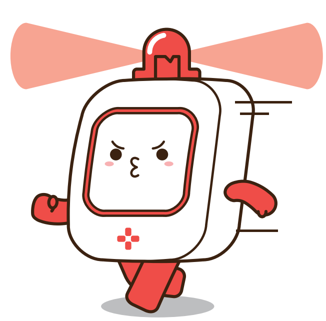
EXAMPLE:



Every time the button is clicked, we want to increment the count of clicks (variable). With the addition block, we can easily track the result. In block coding, when a variable is created, its initial value is 0 unless specified otherwise.

THINK ABOUT IT!

If you want the number to decrease every time the button is clicked, you should change it to subtraction operation. So, you need to modify the code to perform subtraction operation.



Flowcharts & Algorithms

Let's read the flowchart and create the code.

FLOWCHART